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Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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1. Introduction

This laboratory focuses on mesh resolution and electrostatic simulation in a silicon MOS double quantum dot. Starting from the device geometry, the mesh density is analyzed in the quantum regions, then the non-linear Poisson equation is solved to study how the applied gate biases shape the conduction-band landscape.

2. Mesh

2.1 Structure Analysis

The device analyzed in this laboratory is a Double Quantum Dot structure fabricated using Silicon MOS technology. The overall device has the dimensions as reported in the table 2.1.

Axis	Length (nm)
X	353
Y	273
Z	126

Table 2.1: Device Dimensions

The device is divided into different regions determined by the code itself, in order to assign different properties both in terms of physical characteristics and in terms of meshing for the calculation of the simulation. Regarding the mesh sizes, in the code are defined different measures that will be assigned to specific regions:

```
grid_spacing_01      = 1; //nm
grid_spacing_02      = grid_spacing_01*2;
grid_spacing_03      = grid_spacing_02*2;
grid_spacing_04      = grid_spacing_03;
grid_spacing_05      = grid_spacing_03*2;
```

Each of this parameter will indicates the local characteristic mesh size assigned to the points of the regions that will constitute the device. Of course, the higher the mesh size, the higher the elements size and so the lower of the mesh density, resulting in a lower accuracy when it comes to the perform calcualtions.

Each point is defined as:

```
Point(id) = {x, y, z, lc};
```

where the last arguement is the one just mentioned.

By analyzing the code, it is possible to indentify the division of the structure, which is shown in table 2.2 starting from the bottom to the top, with their respective mesh sizes.

As we can see, the regions with the highest mesh density are the ones that correponds to the volume of space where the quantum dots are located, togheter with the regions right above and right below them. This is obvious, since here we need to calculate the poisson equation with the highest accuracy.

Region name	Mesh size (nm)
region_bulk_bottom	8
region_pre_quantum_A/B/C (bottom face)	2
region_pre_quantum_A/B/C (top face)	1
region_quantum_A/B/C	1
region_post_quantum_A/B/C	2
region_bulk_top	2
region_bottom_oxide_layer	4
region_Y_gate	4
region_top_oxide_layer	8
region_Barrier/Plunger/Reservoir_X_gate	4

Table 2.2: Device regions and mesh sizes

3. Poisson

3.1 Simulation setup

The Poisson simulation is defined in `SiMOS_poisson.py`. The device is loaded from the three-dimensional SiMOS mesh and instantiated as an electron device,

```
d = Device(mesh, conf_carriers="e")
d.set_temperature(T)
d.statistics = "FD_approx"
```

therefore the self-consistent charge density is evaluated for confined electrons using a Fermi–Dirac approximation. The operating temperature is

$$T=0.015\text{K}=15\text{mK}.$$

The electrostatic biases are applied only through the surfaces declared with `new_gate_bnd`. The work-function parameter used for all gate boundaries is

$$\Phi_m = \chi_{\text{Si}} + \frac{E_{g,\text{Si}}}{2},$$

which corresponds to a mid-gap silicon reference. The voltages used in the simulation are summarized in Table 3.1.

Boundary	Voltage (V)	Role
surface_Y1_gate	0.535	Longitudinal confinement gate.
surface_Y2_gate	0.535	Longitudinal confinement gate, biased symmetrically with Y1.
surface_Reservoir_L_X_gate	1.300	Left reservoir accumulation.
surface_Reservoir_R_X_gate	1.300	Right reservoir accumulation.
surface_Barrier_L_X_gate	0.040	Left tunnel barrier.
surface_Barrier_M_X_gate	0.400	Inter-dot barrier.
surface_Barrier_R_X_gate	0.040	Right tunnel barrier.
surface_Plunger_L_X_gate	0.700	Left dot plunger.
surface_Plunger_R_X_gate	0.7001990578	Right dot plunger, detuned by about 199 μ V.

Table 3.1: Gate boundary voltages used in the Poisson simulation.

No ohmic boundary is active in the script, since the two corresponding `new_ohmic_bnd` lines are commented. The reservoirs are instead controlled electrostatically by the reservoir gates.

3.2 Materials and regions

The material assignment is performed region by region before the boundary conditions are imposed. The code uses three QTCAD material objects, reported in Table 3.2. The material choice enters the Poisson problem mainly through the dielectric response and, in the semiconductor, through the band parameters and carrier statistics used to compute the charge density.

QTCAD material	Physical material	Use in the device
mt.Si	Silicon	Semiconductor body, reservoirs, and quantum regions. Some silicon regions are n-doped with $N_D = 5 \times 10^{24} \text{m}^{-3}$, corresponding to $5 \times 10^{18} \text{cm}^{-3}$; the p-doping is set to zero.
mt.SiO2	Silicon dioxide	Oxide regions immediately above the quantum layer and in the upper central stack. These regions act as insulating dielectrics and do not contribute mobile carriers.
mt.Al2O3	Aluminum oxide	Top oxide and gate-stack regions. The metallic action of the gates is not represented by assigning a metal material to the volume, but by imposing Dirichlet gate boundary conditions on the gate surfaces.

Table 3.2: Materials appearing in `SiMOS_poisson.py`.

The quantum-dot region used by the solver is

```
dot_region={region_pre_quantum_B_mid,region_quantum_B_mid,region_post_quantum_B_mid}.
```

This central stack is also passed to `d.set_dot_region`, so classical charge is suppressed there. This is consistent with treating the dot as a region where only few-electron quantum states should be described later, rather than as a classical electron gas that screens the gates.

3.3 Poisson solver parameters

The solver is instantiated with a geometry file,

```
p = PoissonSolver(d, solver_params=params_poisson, geo_file=path_geo)
```

so QTCAD uses the adaptive-mesh non-linear Poisson solver. The meaning of each parameter in the script is summarized in Table 3.3.

Parameter	Value	Meaning in this simulation
<code>tol</code>	10^{-7}	Convergence tolerance for the self-consistent Poisson loop. With the default absolute error criterion, the iteration stops when the maximum change in electrostatic potential between successive iterations is below 10^{-7} V.
<code>initial_ref_factor</code>	0.55	Initial adaptive-refinement factor. It is dimensionless; values closer to zero force stronger mesh refinement after non-convergence, while values closer to one refine more conservatively.
<code>final_ref_factor</code>	0.85	Refinement factor used once the solver error has saturated. The high value means the final adaptive refinements are intentionally mild, avoiding an unnecessarily large mesh.
<code>min_nodes</code>	0	No lower bound is imposed on the total number of mesh nodes. If the electrostatic solution converges, the adaptive procedure is allowed to stop even for a relatively small mesh.
<code>maxiter</code>	1000	Maximum number of self-consistent non-linear Poisson iterations before the solve is considered unsuccessful. This is larger than the QTCAD default and is useful at the very low operating temperature.
<code>refined_region</code>	<code>dot_region</code>	Region in which QTCAD enforces a maximum local mesh characteristic length. Here the enforced refinement is focused on the central dot stack.
<code>h_refined</code>	0.3	Maximum characteristic mesh length in the regions listed in <code>refined_region</code> . Since the mesh is loaded with a 10^{-9} scaling factor and the Gmsh geometry is in nanometers, this corresponds to 0.3nm.
<code>max_nodes</code>	10^{10}	Upper bound on the number of mesh nodes allowed during adaptive refinement. This value is effectively non-limiting for the present calculation; the practical bound will be memory and runtime.

Table 3.3: Meaning of the Poisson solver parameters used in the adaptive calculation.

Overall, the parameter set prioritizes a tight electrostatic convergence criterion and high spatial resolution in the quantum-dot region. The adaptive refinement is not globally aggressive: the high refinement factors and the absence of a minimum-node requirement avoid spending nodes in regions where the potential is smooth, while `h_refined` guarantees a fine mesh in the central region that controls the double-dot confinement.